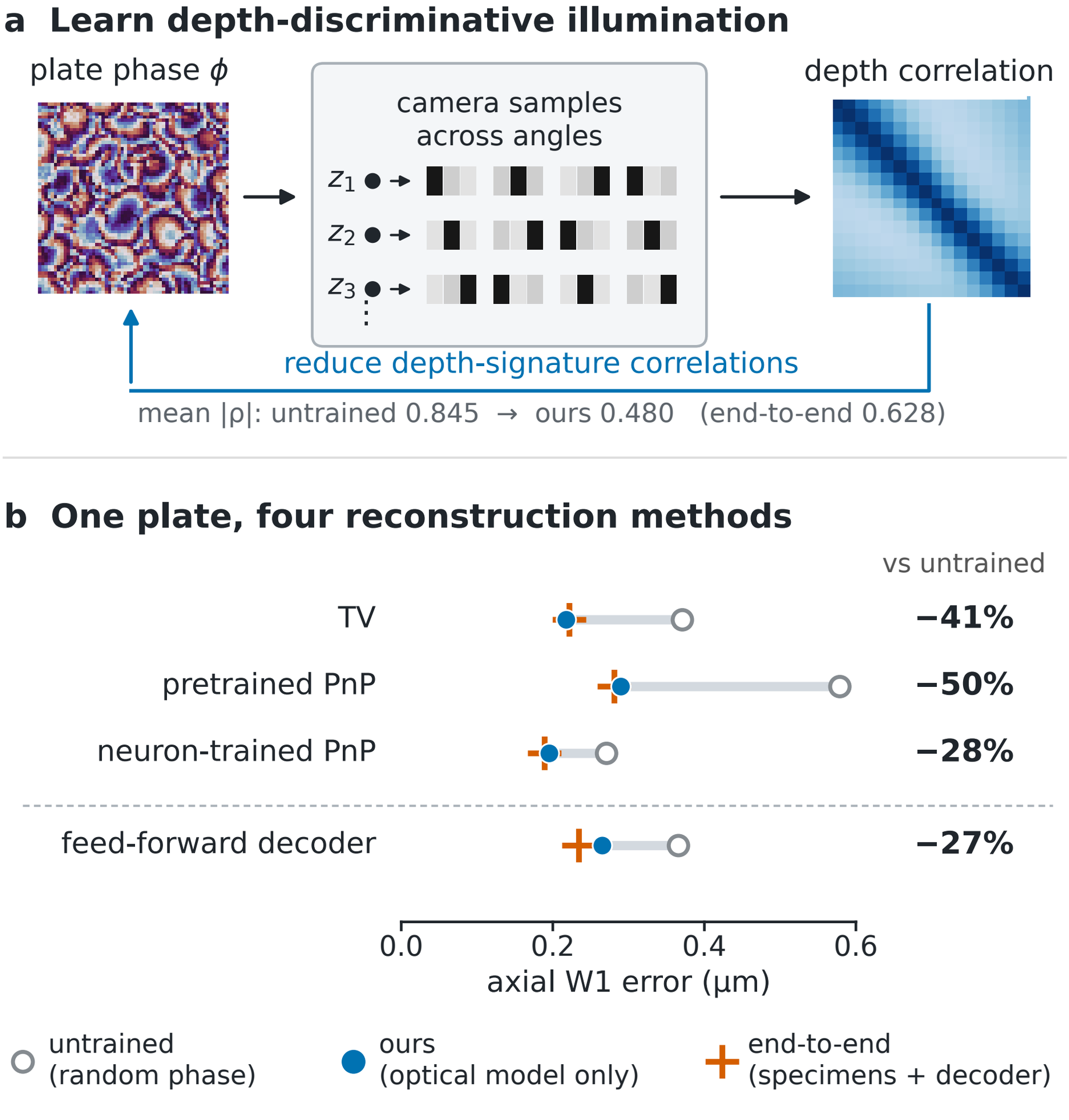
a Learn depth-discriminative illumination
plate phase ϕ
camera samples
across angles
z_1
z_2
z_3
depth correlation
reduce depth-signature correlations
mean |ρ|: untrained 0.845 → ours 0.480 (end-to-end 0.628)
b One plate, four reconstruction methods
vs untrained
TV
−41%
pretrained PnP
−50%
neuron-trained PnP
−28%
feed-forward decoder
−27%
0.0
0.2
0.4
0.6
axial W1 error (μm)
untrained
(random phase)
ours
(optical model only)
end-to-end
(specimens + decoder)